

TABLE II.3
Properties of the elements and certain molecules

Element or molecule	Symbol	Atomic number	Atomic or molecular weight*	Nominal density, g/cm ³	Atoms or molecules per cm ³ † ($\times 10^{24}$)	σ_a , † barns	σ_{sc} , † barns	Σ_a , † cm ⁻¹	Σ_{sc} , † cm ⁻¹
Actinium	Ac	89	227			-515			
Aluminum	Al	13	26.9815	2.699	0.06024	0.230	1.49	0.01386	0.08976
Antimony	Sb	51	121.75	6.62	0.03275	5.4	4.2	0.1769	0.1376
Argon	Ar	18	39.948	Gas		0.678	0.644		
Arsenic	As	33	74.9216	5.73	0.04606	4.3	7	0.1981	0.3224
Barium	Ba	56	137.34	3.5	0.01535	1.2		0.01842	
Beryllium	Be	4	9.0122	1.85	0.1236	0.0092	6.14	0.001137	0.7589
Bismuth	Bi	83	208.980	9.80	0.02824	0.033		0.0009319	
Boron	B	5	10.811	2.3	0.1281	759	3.6	97.23	0.4612
Bromine	Br	35	79.909	3.12	0.02351	6.8	6.1	0.1599	0.1434
Cadmium	Cd	48	112.40	8.65	0.04635	2450	5.6	113.56	0.2596
Calcium	Ca	20	40.08	1.55	0.02329	0.43		0.01001	
Carbon (graphite)§	C	6	12.01115	1.60	0.08023	0.0034	4.75	0.0002728	0.3811
Cerium	Ce	58	140.12	6.78	0.02914	0.63	4.7	0.01836	0.1370
Cesium	Cs	55	132.905	1.9	0.008610	29.0		0.2497	
Chlorine	Cl	17	35.453	Gas		33.2			
Chromium	Cr	24	51.996	7.19	0.08328	3.1	3.8	0.2582	0.3165
Cobalt	Co	27	58.9332	8.8	0.08993	37.2	6.7	3.345	0.6025
Copper	Cu	29	63.54	8.96	0.08493	3.79	7.9	0.3219	0.6709
Deuterium	D	1	2.01410	Gas		0.00053			
Dysprosium	Dy	66	162.50	8.56	0.03172	930	100	29.50	3.172
Erbium	Er	68	167.26	9.16	0.03203	162	11.0	5.189	0.3523
Europium	Eu	63	151.96	5.22	0.02069	4600	8.0	95.17	0.1655
Fluorine	F	9	18.9984	Gas		0.0095	4.0		
Gadolinium	Gd	64	157.25	7.95	0.03045	49000		1492	
Gallium	Ga	31	69.72	5.91	0.05105	2.9	6.5	0.1480	0.3318
Germanium	Ge	32	72.59	5.36	0.04447	2.3	7.5	0.1023	0.3335
Gold	Au	79	196.967	19.32	0.05907	98.8		5.836	
Hafnium	Hf	72	178.49	13.36	0.04508	102	8	4.598	0.3606
Heavy water¶	D ₂ O		20.0276	1.105	0.03323	0.00133	13.6	4.420×10^{-5}	0.4519
Helium	He	2	4.0026	Gas		<0.05			
Holmium	Ho	67	164.930	8.76	0.03199	66.5	9.4	2.127	0.3007
Hydrogen	H	1	1.00797	Gas		0.332			
Indium	In	49	114.82	7.31	0.03834	193.5		7.419	
Iodine	I	53	126.9044	4.93	0.02340	6.2		0.1451	
Iridium	Ir	77	192.2	22.5	0.07050	426	14	30.03	0.9870
Iron	Fe	26	55.847	7.87	0.08487	2.55	10.9	0.2164	0.9251
Krypton	Kr	36	83.80	Gas		25.0	7.50		
Lanthanum	La	57	138.91	6.19	0.02684	9.0	9.3	0.2416	0.2496
Lead	Pb	82	207.19	11.34	0.03296	0.170	11.4	0.005603	0.3757
Lithium	Li	3	6.942	0.53	0.04600	70.7		3.252	
Lutetium	Lu	71	174.97	9.74	0.03353	77	8	2.581	0.2682
Magnesium	Mg	12	24.312	1.74	0.04310	0.063	3.42	0.002715	0.1474
Manganese	Mn	25	54.9380	7.43	0.08145	13.3	2.1	1.083	0.1710
Mercury	Hg	80	200.59	13.55	0.04068	375		15.26	
Molybdenum	Mo	42	95.94	10.2	0.06403	2.65	5.8	0.1697	0.3714
Neodymium	Nd	60	144.24	6.98	0.02914	50.5	16	1.472	0.4662
Neon	Ne	10	20.183	Gas		0.038	2.42		
Nickel	Ni	28	58.71	8.90	0.09130	4.43	17.3	0.4045	1.579
Niobium	Nb	41	92.906	8.57	0.05555	1.15		0.06388	
Nitrogen	N	7	14.0067	Gas		1.85	10.6		
Osmium	Os	76	190.2	22.5	0.07124	15.3		1.090	
Oxygen	O	8	15.9994	Gas		0.00027	3.76		
Palladium	Pd	46	106.4	12.0	0.06792	6.9	5.0	0.4686	0.3396
Protactinium	Pa	91	231.036	1.82	0.03539	0.180		0.006370	

TABLE II.3 (cont.)

Element or molecule	Symbol	Atomic number	Atomic or molecular weight*	Nominal density, g/cm ³	Atoms or molecules per cm ³ † ($\times 10^{24}$)	σ_a , ‡ barns	σ_s , ‡ barns	Σ_a , † cm ⁻¹	Σ_s , † cm ⁻¹
Platinum	Pt	78	195.09	21.45	0.06622	10.0	11.2	0.6622	0.7167
Plutonium	Pu	94	239.0522	19.6	0.04938	$\sigma_a = 1011.3$	7.7	49.93	0.3802
Polonium	Po	84	210	9.51	0.02727	$\sigma_f = 742.5$		36.66	
Potassium	K	19	39.102	0.86	0.01325	2.10	1.5	0.02783	0.01988
Praseodymium	Pr	59	140.907	6.78	0.02898	11.5	3.3	0.3333	0.09563
Promethium	Pm	61							
Protactinium	Pa	91	231.0359			210			
Radium	Ra	88	226.0254	5.0	0.01332	11.5		0.1532	
Rhenium	Re	75	186.2	20	0.06596	88	11.3	5.804	0.7453
Rhodium	Rh	45	102.905	12.41	0.07263	150		10.89	
Rubidium	Rb	37	85.47	1.53	0.01078	0.37	6.2	0.003989	0.06684
Ruthenium	Ru	44	101.07	12.2	0.07270	2.56		0.1861	
Samarium	Sm	62	150.35	6.93	0.02776	5800		161.0	
Scandium	Sc	21	44.956	2.5	0.03349	26.5	24	0.8875	0.8038
Selenium	Se	34	78.96	4.81	0.03669	11.7	9.7	0.4293	0.3559
Silicon	Si	14	28.086	2.33	0.04996	0.16	2.2	0.007994	0.1099
Silver	Ag	47	107.870	10.49	0.05857	63.6		3.725	
Sodium	Na	11	22.9898	0.97	0.02541	0.530	3.2	0.01347	0.08131
Strontium	Sr	38	87.62	2.6	0.01787	1.21	10	0.02162	0.1787
Sulfur (yellow)	S	16	32.064	2.07	0.03888	0.520	0.975	0.02022	0.03791
Tantalum	Ta	73	180.948	16.6	0.05525	21.0	6.2	1.160	0.3426
Technetium	Tc	43	99			19			
Tellurium	Te	52	127.60	6.24	0.02945	4.7		0.1384	
Terbium	Tb	65	158.925	8.33	0.03157	25.5	20	0.8050	0.6314
Thallium	Tl	81	204.37	11.85	0.03492	3.4	9.7	0.1187	0.3387
Thorium	Th	90	232.038	11.71	0.03039	7.40	12.67	0.2249	0.3850
Thulium	Tm	69	168.934	9.35	0.03314	103	12	3.413	0.3977
Tin	Sn	50	118.69	7.298	0.03703	0.63		0.02333	
Titanium	Ti	22	47.90	4.51	0.05670	6.1	4.0	0.3459	0.2268
Tungsten	W	74	183.85	19.2	0.06289	18.5		1.163	
Uranium	U	92	238.03	19.1	0.04833	$\sigma_a = 7.59$ $\sigma_f = 4.19$	8.90	0.3668	0.4301
Vanadium	V	23	50.942	6.1	0.07212	5.04	4.93	0.2025	
Water	H ₂ O		18.0153	1.0	0.03343	0.664	103	0.3635	0.3556
Xenon	Xe	54	131.30	Gas		24.5	4.30	0.02220	3.443
Ytterbium	Yb	70	173.04	7.01	0.02440	36.6	25.0	0.8930	0.6100
Yttrium	Y	39	88.906	5.51	0.03733	1.28	7.60	0.04778	0.2837
Zinc	Zn	30	65.37	7.133	0.06572	1.10	4.2	0.07230	0.2760
Zirconium	Zr	40	91.22	6.5	0.04291	0.185	6.40	0.007938	0.2746

*Based on ¹²C = 12.00000.

†Four-digit accuracy for computational purposes only; last digit(s) usually is not meaningful.

‡Cross sections at 0.0253 eV or 2200 m/sec. The scattering cross sections, except for those of H₂O and D₂O, are measured values in a thermal neutron spectrum and are assumed to be 0.0253 eV values because σ_s is usually constant at thermal energies. The errors in σ_s tend to be large, and the tabulated values of σ_s should be used with caution. (From BNL-325, 3rd ed., 1973).§The value of σ_a given in the table is for pure graphite. Commercial, reactor-grade graphite contains varying amounts of contaminants and σ_a is somewhat larger, say, about 0.0048 barns, so that $\Sigma_a \approx 0.0003851$ cm⁻¹.¶The value of σ_a given in the table is for pure D₂O. Commercially available heavy water contains small amounts of ordinary water and σ_a in this case is somewhat larger.